

Project Demonstration

Date	11 July 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of the **Voice-Based Concept Understanding Analyzer**. The application accepts a student's voice explanation, converts speech into text, performs semantic analysis, evaluates conceptual understanding, and generates a detailed PDF report.

1. System Architecture

DATA FLOW DIAGRAM (LEVEL 0)

Voice-Based Concept Understanding Analyzer

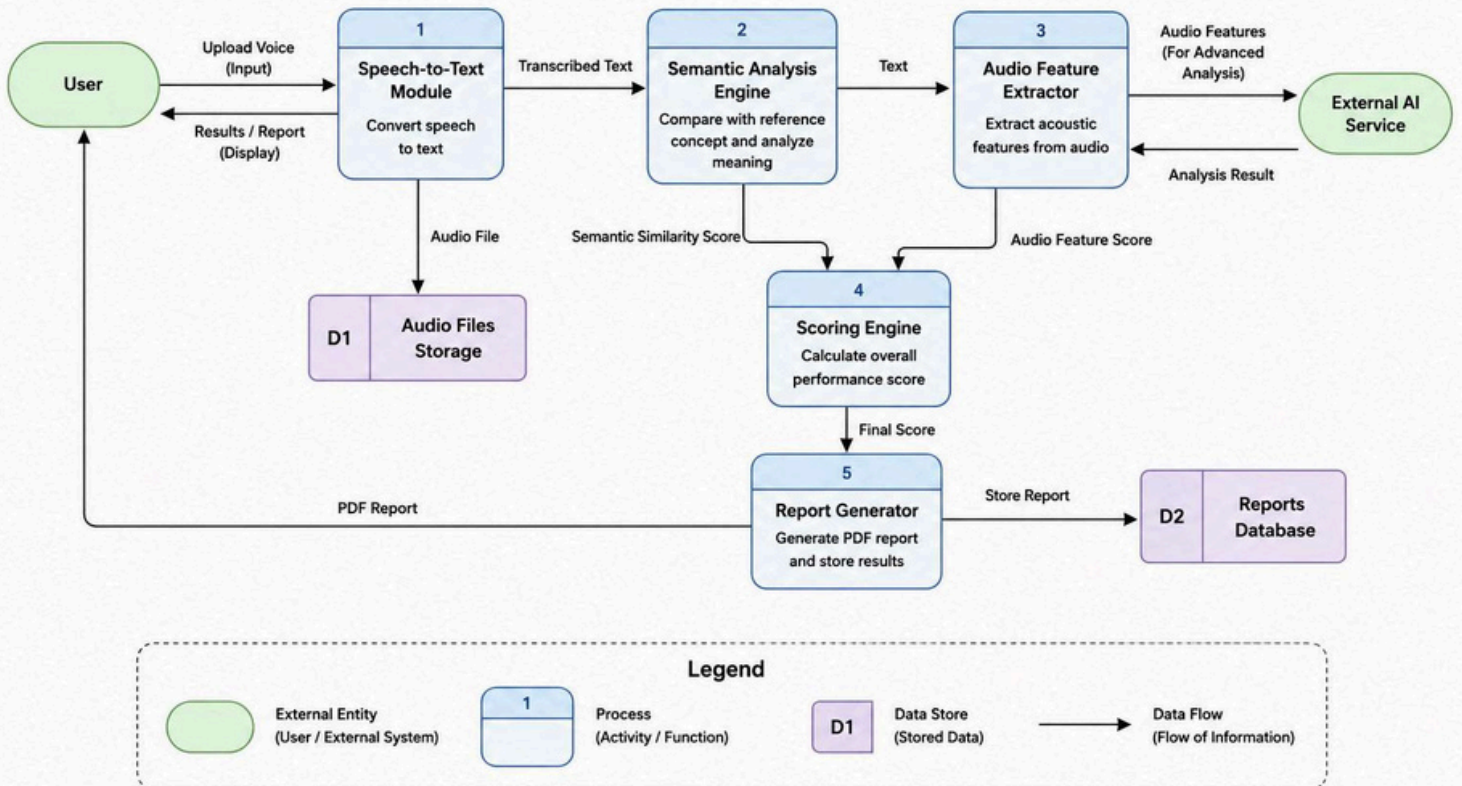
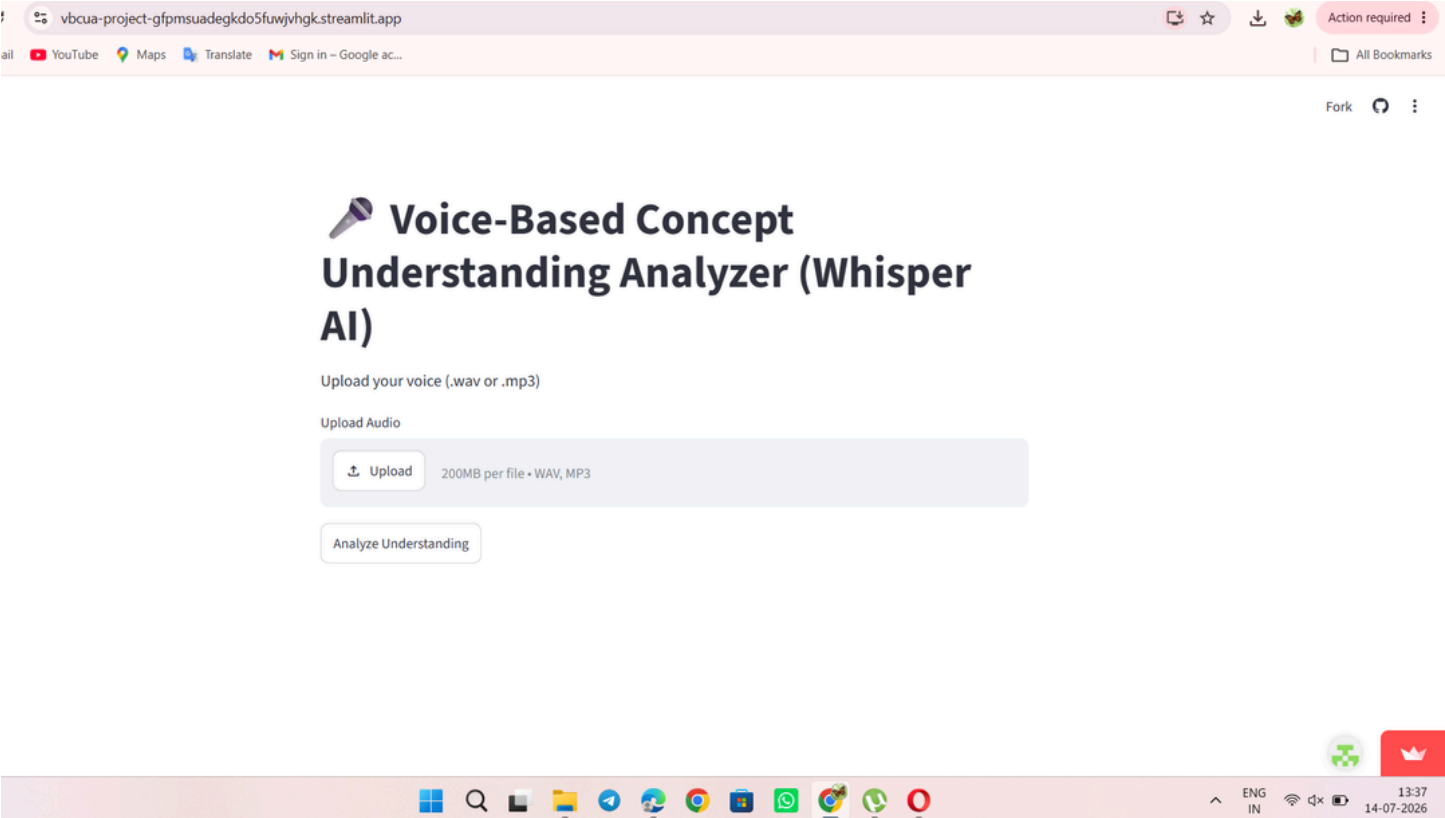


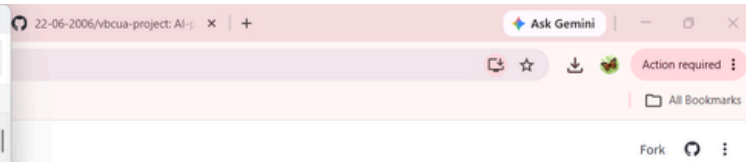
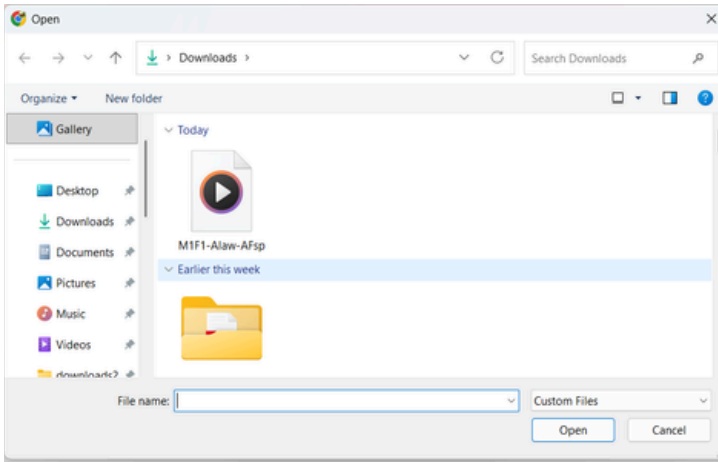
Figure 1: Data Flow Diagram

2. Project Structure

```
VBCUA-Project/
├── app.py
├── requirements.txt
├── packages.txt
├── README.md
├── LICENSE
├── .gitignore
├── assets/
│   ├── screenshots/
│   └── logo.png
├── audio/
│   └── sample_audio.wav
├── models/
├── utils/
│   ├── speech_to_text.py
│   ├── semantic_analysis.py
│   ├── scoring.py
│   └── pdf_generator.py
├── reports/
│   └── sample_report.pdf
└── temp/
    └── temp_audio.wav
```

3 . Streamlit Application Running





Concept alyzer (Whisper)

Upload 200MB per file • WAV, MP3

Analyze Understanding

IRFC
-1.21%



ENG
IN



13:37
14-07-2026



Voice-Based Concept Understanding Analyzer (Whisper AI)

Upload your voice (.wav or .mp3)

Upload Audio



M1F1-Alaw-AFsp.wav

46.1KB

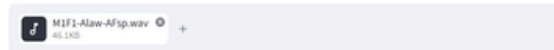
+

Analyze Understanding

Voice-Based Concept Understanding Analyzer (Whisper AI)

Upload your voice (.wav or .mp3)

Upload Audio



Transcribed Text:

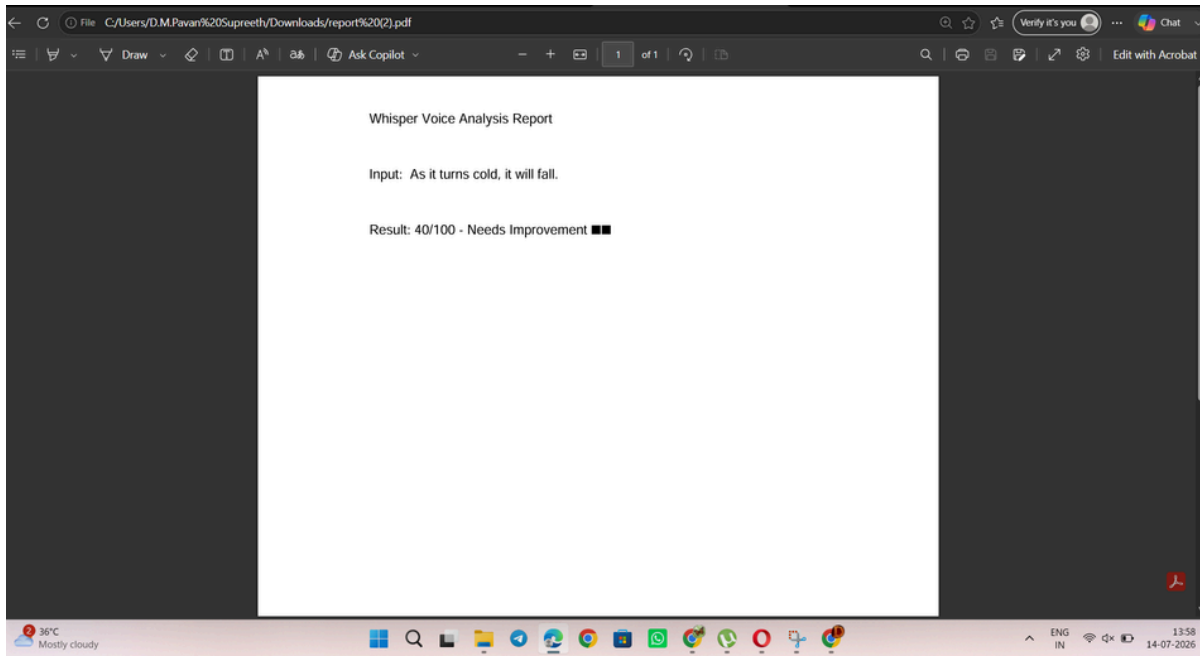
As it turns cold, it will fall.

Analyze Understanding

40/100 - Needs Improvement 🟡

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4. OUTPUT :



5. Demonstration Summary

The application was successfully executed and tested. All major functionalities including audio upload, speech-to-text conversion, semantic similarity analysis, score generation, waveform visualization, dashboard display, and PDF report generation worked successfully during the demonstration.